Learning to Price Political Risk: Stock Market Volatility Responses to U.S.–China Trade Policy Shocks Across Two Trump Administrations

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This paper compares U.S. and Chinese equity market reactions to trade policy announcements targeting China across Trump's two presidential terms (2017–2019 and 2025–2026). Using tick-level data from the NYSE TAQ Millisecond database for 12 U.S. ETFs and 5-minute bar data from the RESSET High-Frequency database for 7 Chinese ETFs, together with 44 trade policy events spanning both administrations, I construct realized volatility measures and implement intraday event studies with permutation-based inference. I find that U.S. intraday abnormal returns around announcements are approximately three times larger in the second term than in the first: mean absolute cumulative abnormal returns in a two-hour window are 1.15% in Term 2 versus 0.39% in Term 1 ($p < 0.01$). For Chinese markets, the pattern reverses: the CSI 300 ETF's mean absolute abnormal return declines from 1.61% in Term 1 to 1.13% in Term 2. A cross-market DID yields a 1.29 percentage point divergence at the event-day level ($p = 0.040$). HAR-RV interaction coefficients are consistently positive for U.S. sectors but individually insignificant; with only 18 second-term events, these tests are underpowered. For Chinese ETFs, the interaction is negative and insignificant, consistent with the declining reaction. Sector-level China revenue exposure does not predict differential amplification. The results point to broad market uncertainty rather than sector-specific trade exposure channels, consistent with belief revision following tail policy realizations that fell outside investors' prior experience. Excluding 5 FOMC/NFP overlap events strengthens the DID to 1.86 pp (p = 0.001).

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1. Introduction

Between 2017 and 2019, the Trump administration raised tariffs on Chinese imports from nearly zero to 25% on roughly $250 billion of goods. The escalation was gradual, proceeding through multiple rounds of Section 301 actions, and it culminated in the Phase One trade agreement of January 2020. When Trump returned to office in January 2025, tariffs on China resumed, but at a pace and scale far exceeding anything in the first term. Within months, announced rates reached 145%, and reciprocal tariffs extended to virtually all trading partners. The scale was unprecedented. This paper examines whether U.S. equity markets responded differently to trade policy shocks in the second term compared to the first, and if so, through what channels.

Using tick-level data from the NYSE TAQ Millisecond database for 12 U.S. ETFs and 5-minute bar data from the RESSET High-Frequency database for 7 Chinese ETFs, together with 44 trade policy events, I find that U.S. intraday abnormal returns around announcements are approximately three times larger in the second term than in the first. In a two-hour window centered on each event, the mean absolute cumulative abnormal return is 1.15% in Term 2 versus 0.39% in Term 1, a difference significant at the 1% level. The result holds across alternative window specifications and test statistics, and it holds across sectors. Markets did not habituate to repeated trade policy shocks; they reacted more.

For Chinese markets, the pattern reverses. The CSI 300 ETF's mean absolute event-day abnormal return declines from 1.61% in Term 1 to 1.13% in Term 2, and the Chinese HAR-RV event×Term 2 interaction is negative for all seven Chinese ETFs, though none reach significance. The cross-market DID combining U.S. and Chinese reactions yields a 1.29 percentage point divergence at the event-day level ($p = 0.040$). U.S. markets amplified; Chinese markets attenuated. The divergence supports differential learning: U.S. investors, having formed posteriors around moderate outcomes after the Phase One agreement, were surprised by the second term's escalation; Chinese investors, operating in a longer history of bilateral trade frictions, had already incorporated the possibility of extreme outcomes.[1]

[1] Pre-Trump U.S.–China trade frictions include the contentious WTO accession negotiations (1999–2001), Section 421 safeguard tariffs on Chinese tires (2009), antidumping and countervailing duty cases on solar panels (2012) and steel products (multiple years), and recurring congressional debates over currency manipulation designation (2003–2015). These episodes, while smaller in scale than the Trump-era tariffs, provided Chinese market participants with repeated signals that bilateral trade policy could shift abruptly.

The two-term comparison is useful as a research design because it holds constant the policy actor, the target country, and the primary instrument (tariffs), while varying the magnitude and sequencing of shocks. Specifically, this provides a cleaner identification strategy than cross-country or cross-episode comparisons, where institutional differences confound inference. The six-fold increase in peak tariff rates between terms provides substantial variation in policy intensity, and the millisecond-resolution trade data permits identification of market reactions within minutes of an announcement.

I construct daily realized volatility following Andersen, Bollerslev, Diebold, and Labys (2003), along with bipower variation and jump components from Barndorff-Nielsen and Shephard (2004, 2006), using tick data aggregated to 5-minute bars. The empirical analysis has four components: an intraday event study at the 5-minute frequency with Boehmer, Musumeci, and Poulsen (1991) cross-sectional tests, Bremer and Zhang (2007) GARCH-adjusted extensions, and permutation-based inference; a HAR-RV model (Corsi, 2009) augmented with event dummies and term interactions for both U.S. and Chinese ETFs; a cross-market DID pairing SPY and the CSI 300 ETF to separate U.S. amplification from Chinese attenuation; and a sectoral panel regression that tests whether China revenue exposure amplifies the response.

The U.S. HAR-RV interaction coefficients are positive for 8 of 10 sector ETFs, but none reach conventional significance levels. Indeed, this is a power limitation, not a contradiction: with only 18 second-term events, the HAR-RV tests are underpowered and I treat them as suggestive. For Chinese ETFs, the event×Term 2 interaction is negative for all seven tickers (CSI 300: $\hat{\delta}_3 = -0.24$, $t = -0.56$; SSE 50: $\hat{\delta}_3 = -0.16$, $t = -0.34$), consistent with declining Chinese reactions. The amplification, however, shows up in directional price moves (cumulative abnormal returns) rather than in the dispersion of intraday returns (realized volatility), which is consistent with markets repricing sharply in one direction rather than whipsawing. The sectoral triple interaction (Event × ChinaExposure × Term2) is statistically indistinguishable from zero ($p = 0.49$), though ex-post power is only 10.2%. Together, these results point to broad market uncertainty rather than sector-specific trade exposure channels. This interpretation is consistent with models of belief revision following tail realizations, particularly Kozlowski, Veldkamp, and Venkateswaran (2020), who formalize how extreme outcomes permanently shift the perceived probability of future extremes.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature and states the contribution. Section 3 describes the econometric methodology. Section 4 discusses data construction and event identification. Section 5 presents the results, including intraday event studies, the cross-market DID, HAR-RV models, sectoral analysis, and directional decomposition. Section 6 discusses interpretations, alternative explanations, and limitations. Section 7 concludes.

2. Related Literature

There is by now a substantial literature on the economic effects of trade policy uncertainty, though relatively little of it examines high-frequency market reactions. In fact, the theoretical foundations are well established. Handley and Limao (2017, 2022) show that uncertain future tariff rates induce firms to delay irreversible investment decisions, which reduces firm value and increases equity risk premia through a real-options channel. Caldara, Iacoviello, Molligo, Prestipino, and Raffo (2020) demonstrate that trade policy uncertainty shocks reduce investment and output even in the absence of actual tariff changes, while Egger and Zhu (2020) document similar stock-market effects using the U.S.–China trade war as a natural experiment. The welfare costs are substantial. Fajgelbaum, Goldberg, Kennedy, and Khandelwal (2020) estimate aggregate annual real income losses of $7.2 billion from the 2018 tariffs alone, and Amiti, Redding, and Weinstein (2019) find that the cost incidence fell almost entirely on U.S. importers rather than Chinese exporters. These estimates imply a direct channel from trade policy events to equity valuations through expected cash flows.

However, the existing empirical work on equity market reactions to the first Trump term relies almost exclusively on daily data. Huang, Lin, Liu, and Tang (2023) find that firms with higher China revenue exposure experienced larger abnormal returns around Section 301 announcements, and Wagner, Zeckhauser, and Ziegler (2018) document lower abnormal returns for internationally exposed firms relative to domestically oriented firms following the 2016 election. Greenland, Ion, Lopresti, and Schott (2020) use equity market reactions to infer trade exposure at the firm level. No existing study, to my knowledge, compares high-frequency market reactions across the two Trump terms, and no study applies realized volatility methods to trade policy events specifically.

The realized volatility measures used in this paper draw on the framework developed by Andersen, Bollerslev, Diebold, and Labys (2001, 2003) and Barndorff-Nielsen and Shephard (2002, 2004, 2006). The key insight is that the sum of squared intraday returns converges to integrated variance as the sampling frequency increases, providing a model-free volatility measure that does not require specifying a parametric model for the variance process. Bipower variation, introduced by Barndorff-Nielsen and Shephard (2004), is consistent for the continuous component of price variation even in the presence of jumps, and the associated z-statistic provides a formal test for jump detection. For modeling volatility dynamics, I rely on Corsi's (2009) Heterogeneous Autoregressive model of Realized Volatility (HAR-RV), which captures the well-documented long-memory properties of volatility through daily, weekly, and monthly components; Patton and Sheppard (2015) and Bollerslev, Patton, and Quaedvlieg (2016) extend this framework in various directions. In terms of event identification, the closest antecedent is the macro-announcement literature. Ederington and Lee (1993) show that most of the price adjustment to scheduled macroeconomic news occurs within the first few minutes, and Beber and Brandt (2009) document sharp declines in implied volatility as uncertainty resolves around announcements. Trade policy announcements differ from scheduled macro releases in being irregular, partially anticipated, and often conveyed through informal channels such as social media posts, which motivates the intraday approach adopted in this paper.

I interpret the two-term comparison primarily through models of political risk and belief revision. Pastor and Veronesi (2012, 2013) develop models in which policy changes command a risk premium because political risk cannot be hedged through standard financial instruments. Kelly, Pastor, and Veronesi (2016) provide empirical confirmation using option prices around elections and major political events. In general, Baker, Bloom, and Davis (2016) document the broader association between economic policy uncertainty and market volatility, and Brogaard and Detzel (2015) show that their EPU index forecasts lower future equity returns. The central question for this paper, whether repeated exposure to trade policy shocks produces habituation or amplification, connects to the literature on learning and experience effects. Collin-Dufresne, Johannes, and Lochstoer (2016) show that parameter learning in equilibrium models generates time-varying risk premia. Malmendier and Nagel (2011) find that lived macroeconomic experiences shape individual risk attitudes persistently, sometimes for decades. The most directly relevant paper for the present study is Kozlowski, Veldkamp, and Venkateswaran (2020), who formalize how the experience of a single tail event can permanently shift beliefs about the probability of future extreme outcomes. In the context of U.S.–China trade policy, the first term's resolution through the Phase One agreement may have tightened investor posteriors around moderate outcomes; the second term's roughly 145% effective tariff rates then represent the kind of tail realization that, in their framework, widens those posteriors and generates persistent increases in perceived risk.

This paper contributes along three dimensions. First, it provides the first cross-term comparison of high-frequency equity market reactions to trade policy shocks from the same policy actor targeting the same bilateral partner, exploiting the six-fold increase in peak tariff rates between terms as a source of variation in policy intensity. Second, it applies realized volatility and jump-detection methods to trade policy events at 5-minute resolution, whereas the existing literature operates at daily frequency. Third, the cross-market DID design separates U.S. amplification from Chinese attenuation, providing evidence on learning and adaptation that a single-market study cannot deliver.

A related literature documents the role of China's "National Team" (国家队), the constellation of state-backed funds that directly purchase equities during market stress. Dang, Gorton, Holmstrom, and Ordonez (2023) find that National Team intervention during the 2015 crash reduced volatility in targeted stocks but simultaneously reduced price informativeness, as government buying crowded out private information acquisition. Brunnermeier, Sockin, and Xiong (2022) provide a theoretical framework in which government stabilization can dampen volatility while diverting investor attention from fundamentals. On the empirical side, Cheng, Guo, Hu, and Zheng (2022) document that National Team holdings lowered crash risk in the short run, but the stabilizing effect disappeared within six months. Li, Zhang, and Zhao (2024) find increased daily volatility but decreased crash risk in National Team stocks, and their evidence suggests that intervention smooths tails without eliminating routine price variation. These findings are, in fact, directly relevant for interpreting the Chinese attenuation result in this paper: state stabilization capacity is real, empirically documented, and potentially confounds the learning interpretation. Section 6 discusses this channel explicitly.

3. Methodology

3.1 Realized Volatility Measures

Let $r_{t,i} = p_{t,i} - p_{t,i-1}$ denote the i-th intraday log return on day t, with $M = 78$ five-minute intervals per trading day (9:30–16:00 ET). Daily realized volatility,

$$RV_t = \sum_{i=1}^{M} r_{t,i}^2,$$

converges in probability to the integrated variance $\int_0^1 \sigma_t^2\,(s)\,ds$ as $M \to \infty$ (Andersen et al., 2003). Bipower variation,

$$BV_t = \frac{\pi}{2} \cdot \frac{M}{M-1} \sum_{i=2}^{M} |\,r_{t,i}| \cdot |r_{t,i-1}|,$$

is consistent for integrated variance even with jumps, because the probability that a jump affects two consecutive returns vanishes as M grows (Barndorff-Nielsen and Shephard, 2004). The jump component is $J_t = \max(RV_t - BV_t,\, 0)$, with continuous component $C_t = BV_t$. I test for jumps using the Barndorff-Nielsen and Shephard (2006) statistic

$$z_{BNS,t} = \frac{RV_t - BV_t}{\sqrt{\theta \cdot M^{-1} \cdot BV_t^2}},$$

where $\theta = (\pi/2)^2 + \pi - 5 \approx 0.6090$. Under the null of no jumps, $z_{BNS,t} \xrightarrow{d} N(0,1)$; I reject at the 5% level when $z_{BNS,t} > 1.645$.

3.2 HAR-RV Model

To model volatility dynamics, I adopt the Heterogeneous Autoregressive model of Realized Volatility (HAR-RV) developed by Corsi (2009), which captures the well-documented persistence in volatility through daily, weekly ($\overline{RV}_{t-1}^{(w)} = \frac{1}{5}\sum_{j=1}^{5} \log RV_{t-j}$), and monthly ($\overline{RV}_{t-1}^{(m)} = \frac{1}{22}\sum_{j=1}^{22} \log RV_{t-j}$) moving averages. I augment the baseline specification with event dummies and term interactions:

$$\log RV_t = \beta_0 + \beta_d \log RV_{t-1} + \beta_w \log \overline{RV}_{t-1}^{(w)} +$$

$$\beta_m \log \overline{RV}_{t-1}^{(m)} + \delta_1 \cdot Event_t + \delta_2 \cdot Term2_t + \delta_3 \cdot (Event_t \times Term2_t) + \gamma \cdot \log VIX_t + \varepsilon_t$$

The three persistence coefficients capture daily, weekly, and monthly volatility components, respectively. The parameter of interest is δ_3, which captures the differential effect of trade policy events on log realized volatility in the second term, after controlling for volatility persistence and market-wide fear as proxied by the VIX. I estimate the model by OLS with Newey-West heteroskedasticity and autocorrelation consistent (HAC) standard errors, with the bandwidth parameter $L = \lfloor 4(n/100)^{2/9} \rfloor$ selected following Andrews (1991). To be more specific, the same specification is estimated for each of the 7 Chinese ETFs, substituting the China Economic Policy Uncertainty index (Baker, Bloom, and Davis, 2016) for the VIX. The China EPU index is available at monthly frequency; I interpolate it to daily using linear interpolation, which smooths within-month variation but is unlikely to bias the event-day coefficients since trade policy events are spaced across months.

3.3 Daily Event Study

For each ticker i and event k, I estimate a market model $R_{i,t} = \alpha_i + \beta_i R_{bench,t} + \epsilon_{i,t}$ over an estimation window of 60 trading days ending 10 days before the event. I use the MSCI World ex-USA index (ACWX) as the benchmark, rather than a U.S. broad market index, in order to avoid including the sector ETFs under study in their own benchmark. Nonetheless, ACWX contains approximately 5–12% Chinese equity weight depending on the period, which means the benchmark is not fully exogenous to U.S.–China trade shocks. This concern is mitigated by the small weight and by the fact that ACWX is used only for U.S. ETF abnormal returns, not Chinese ones. As a robustness check, results are qualitatively unchanged when using SPY itself as the benchmark for sector ETFs (excluding the sector under study). Abnormal returns are then $AR_{i,t} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{bench,t})$, and I cumulate them over event windows $[\tau_1, \tau_2]$ in the standard way. For Chinese ETFs, I use a mean-adjusted abnormal return model, $AR_{i,t} = R_{i,t} - \bar{R}_i$, where $\bar{R}_i$ is the mean return estimated over the same 60-day window. This avoids the need for a Chinese benchmark index that may itself be affected by the same trade policy events. In addition to the standard BMP test, I report the Bremer and Zhang (2007) GARCH-based extension, which accounts for time-varying volatility in the standardization of abnormal returns. This correction is relevant for trade policy events, where event-day variance is unlikely to be constant across events; the standard BMP test can lose power when this assumption is violated.

3.4 Intraday Event Study

Because intraday returns exhibit strong time-of-day patterns (volatility is highest at the open and close, and lowest around midday), the intraday event study must control for these effects. I do so by estimating expected returns $\hat{\mu}_s$ and standard deviations $\hat{\sigma}_s$ for each 5-minute slot s from a rolling 60-day estimation window ending 5 days before each event. The abnormal return at offset ℓ from event bar j is then $AR_{j+\ell} = r_{t,j+\ell} - \hat{\mu}_{s(j+\ell)}$, and I standardize it as $SAR_{j+\ell} = AR_{j+\ell}/\hat{\sigma}_{s(j+\ell)}$. Cumulative abnormal returns over window $[\ell_1, \ell_2]$ are computed as $CAR[\ell_1, \ell_2] = \sum_{\ell=\ell_1}^{\ell_2} AR_{j+\ell}$.

For cross-sectional inference across K events within a given term, I use the Boehmer, Musumeci, and Poulsen (1991) statistic $t_{BMP} = \overline{SCAR}/(\hat{\sigma}_{SCAR}/\sqrt{K})$, which adjusts for event-induced variance—an important consideration since trade policy events are likely to increase return variance on the announcement day itself. To compare reaction magnitudes across the two terms, I test whether the distributions of $|CAR|$ differ using both a Wilcoxon rank-sum test and a permutation test with 10,000 replications. The permutation test is particularly useful here because it makes no distributional assumptions and provides exact inference under the null, which matters given the relatively small number of events in each term.

3.4.1 Time Zone Mapping for Chinese Events

Because U.S. and Chinese trading hours do not overlap, mapping event timing to the appropriate Chinese trading session requires care. U.S. Eastern Time is 12 or 13 hours behind China Standard Time depending on daylight saving. I assign each event to a Chinese market reaction day as follows: events occurring before 03:00 ET on day d (i.e., before the Chinese close at 15:00 CST on day d) are mapped to Chinese trading day d; events occurring at or after 03:00 ET on day d are mapped to the next Chinese trading day. This rule reflects the fact that events announced during U.S. trading hours on day d will first be priced in the Chinese market on day $d + 1$ (or the next trading day if $d + 1$ is a non-trading day). Weekend or holiday events are mapped to the next Chinese trading day. Because multiple U.S. events can map to the same

Chinese trading day, and some events coincide with Chinese market holidays, the 44 U.S. events yield 26 unique Chinese event-days in Term 1 and 12 in Term 2 for the daily analysis (11 for the intraday analysis, which requires complete 5-minute data on the event day). The resulting Chinese event dates are used consistently across all Chinese analyses.

3.5 Sectoral Analysis

If the amplification is driven by sector-specific trade exposure rather than broad uncertainty, sectors with greater revenue dependence on China should react more strongly. To test this, I estimate a panel regression of $\log RV_{i,t}$ for U.S. sector ETFs on a lagged dependent variable, event dummies, sector-level China revenue exposure ($Exposure_i$, approximated from Compustat geographic segment data following the methodology of Huang, Lin, Liu, and Tang, 2023), and all relevant two- and three-way interactions:

$$
\begin{aligned}
&\log RV_{i,t} \\
&= \alpha + \beta_1 \log RV_{i,t-1} + \delta_1 Event_t + \delta_2 Exposure_i + \delta_3 (Event_t \times Exposure_i) + \delta_4 Term2_t \\
&+ \delta_5 (Event_t \times Term2_t) + \delta_6 (Exposure_i \times Term2_t) \\
&+ \delta_7 (Event_t \times Exposure_i \times Term2_t) + \varepsilon_{i,t}
\end{aligned}
$$

The coefficient of interest is δ_7. A positive and significant estimate would indicate that the second term's amplification is concentrated in sectors with high China exposure; an insignificant estimate would point instead toward a broad market uncertainty channel.

3.6 Cross-Market DID

To compare U.S. and Chinese market reactions across terms, I estimate a cross-market DID using absolute cumulative abnormal returns: $DID = (\overline{|CAR|}_{US,T2} - \overline{|CAR|}_{US,T1}) -$

($\overline{|CAR|}_{CN,T2} - \overline{|CAR|}_{CN,T1}$). Inference relies on a permutation test (10,000 replications) that randomly reassigns term labels within each market, and on a regression specification $|CAR|_{i,k} = \alpha + \beta_1 US_i + \beta_2 Term2_k + \beta_3(US_i \times Term2_k) + \varepsilon_{i,k}$ with event-clustered standard errors. I also estimate this with event fixed effects, which absorb β_2 and all event-specific heterogeneity. The use of $|CAR|$ rather than signed returns means that β_3 estimates a difference in reaction magnitudes, not a directional treatment effect in the standard DID sense. Section 5.7 complements this with signed-return analysis by event direction.

3.7 Robustness

The choice of 5-minute sampling frequency is validated using signature plots, which track realized volatility as a function of sampling interval from 1 minute to 325 minutes. If microstructure noise dominates at high frequencies, RV will rise sharply as the interval shrinks below some threshold; conversely, if the interval is too coarse, RV loses information about intraday price variation. The signature plots for SPY, QQQ, and all sector ETFs show that RV stabilizes by 5 minutes and remains flat through 15 minutes, which confirms that the 5-minute frequency balances noise reduction against information loss. The Chinese ETFs show a similar stabilization pattern starting at 5 minutes.

The intraday event study results are reported across multiple window specifications (±6, ±12, ±30, ±60, and full-day), so that the reader can assess whether the cross-term differences are sensitive to horizon choice. In practice, the amplification finding is strongest at the narrowest windows and attenuates at wider horizons, consistent with a trade-specific information shock that is diluted by non-event noise at longer intervals.

I also implement a leave-one-out sensitivity check for the cross-market DID. Each event is dropped in turn, and the permutation test is re-run on the remaining sample. Indeed, the AR[0] result survives the exclusion of every individual event, which reduces the concern that one or two extreme observations drive the cross-market divergence.

The Bremer and Zhang (2007) GARCH-adjusted test statistic complements the standard BMP test by accounting for event-induced variance. In event studies where announcement days are expected to be noisier than the estimation window, the standard cross-sectional test will be oversized. The Bremer-Zhang correction scales each event's abnormal return by a conditional variance estimate, and Specifically, I report both statistics side by side in Table 5 so the reader can assess where the correction matters. The permutation test provides an additional layer of distribution-free inference that is robust to non-normality, serial correlation, and cross-sectional dependence in returns.

4. Data

4.1 U.S. Intraday Data

Tick-level trade data from the NYSE TAQ Millisecond database (via WRDS) are aggregated to 5-minute bars for 12 ETFs using standard quality filters (regular trades, positive price and size, no special conditions). The sample covers Term 1 (July 2016–December 2019, 881 trading days) and Term 2 (July 2024–March 2026, 432 trading days), with each period beginning six months before the first event to provide estimation windows. The 12 ETFs are SPY, QQQ, and 10 sector SPDRs; XLC begins in June 2018. Table 1 summarizes the data. The total sample contains approximately 1.1 million 5-minute observations.

Table 1: Intraday Data Summary

Ticker	Sector	China Exp.	Days T1	Days T2	Total 5-min Bars
SPY	S&P 500	—	881	432	102,414
QQQ	Nasdaq-100	0.22	881	432	102,414
XLK	Technology	0.25	880	432	102,336
XLB	Materials	0.15	881	432	102,414
XLI	Industrials	0.12	880	432	102,336
XLY	Consumer Discretionary	0.10	880	432	102,336
XLE	Energy	0.08	881	432	102,414
XLV	Healthcare	0.07	880	432	102,336
XLC	Communication Services	0.06	387	432	63,882
XLF	Financials	0.03	880	432	102,336
XLP	Consumer Staples	0.05	880	—	68,640
XLU	Utilities	0.01	880	—	68,640

Note: Data from NYSE TAQ Millisecond database (via WRDS), aggregated to 5-minute bars. ETFs: SPY, QQQ, and 10 sector SPDRs (XLB, XLC, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY). XLC begins June 2018. Term 1: July 2016–December 2019 (881 trading days). Term 2: July 2024–March 2026 (432 trading days). Standard quality filters applied (regular trades, positive price and size, no special conditions).

4.2 Chinese Intraday Data

Five-minute bar data for 7 Chinese ETFs are obtained from the RESSET High-Frequency database. The Chinese ETFs are selected to span broad market indices and key sectors affected by trade policy: CSI 300 ETF (510300), SSE 50 ETF (510050), ChiNext ETF (159915), Medical ETF (512010), Securities ETF (512880), Bank ETF (512800), and Consumer ETF (159928). The Chinese trading day consists of a morning session (9:30–11:30 CST) and an afternoon session (13:00–15:00 CST), yielding 48 five-minute bars per day. Table 2 summarizes the Chinese data.

Table 2: Chinese Intraday Data Summary

Ticker	Name	Sector	Days T1	Days T2
510300	CSI 300 ETF	Broad Market	851	179
510050	SSE 50 ETF	Large Cap	851	179
159915	ChiNext ETF	Growth/Tech	851	179

512010	Medical ETF	Healthcare	851	179
512880	Securities ETF	Financials	851	179
512800	Bank ETF	Banking	851	179
159928	Consumer ETF	Consumer	851	179

Note: Data from RESSET High-Frequency database, aggregated to 5-minute bars. ETFs: CSI 300 (510300), SSE 50 (510050), ChiNext (159915), Medical (512010), Securities (512880), Bank (512800), Consumer (159928). Trading hours: 9:30–11:30 and 13:00–15:00 CST (48 bars per day). Sample periods match Table 1.

4.3 Event Identification

I identify 44 trade policy events: 26 in Term 1 (August 2017–December 2019) and 18 in Term 2 (February 2025–March 2026). Events are drawn from official government sources including the White House, the Office of the U.S. Trade Representative, the Federal Register, and the Chinese Ministry of Commerce (MOFCOM). Each event is classified along three dimensions: direction (escalation or de-escalation), initiator (U.S. or China), and announcement timestamp in Eastern Time.

Because the event set is manually curated rather than algorithmically generated, transparent inclusion criteria are essential. An announcement is included if it satisfies all six of the following conditions:

1. *Official policy action.* The event must reflect a concrete government action—an executive order, proclamation, tariff implementation, formal investigation, or binding agreement—rather than speculation, leaked briefings, or unofficial statements.
2. *Bilateral specificity.* The action must directly target U.S.–China bilateral trade. Actions of general scope are included only if they have a first-order effect on bilateral flows. The March 2018 steel and aluminum tariffs, for example, are included because China was one of the principal targets and the announcement was widely understood in market commentary as directed at China, even though the tariffs applied globally.
3. *Material tariff or trade policy change.* The event must involve a tariff rate change, a new trade restriction, or a formal reversal of an existing restriction. Routine administrative notices, minor technical corrections, and product-level exclusion decisions are excluded.
4. *Identifiable announcement date.* The event must have a verifiable date on which the policy action was publicly announced or took effect. Events that unfolded gradually without a discrete announcement are excluded.
5. *Non-overlap.* If multiple related actions occur on the same calendar day, they are consolidated into a single event to avoid double-counting, unless the actions move in opposite directions (e.g., a de-escalation for most

countries paired with an escalation targeting China specifically), in which case both are retained because they carry distinct information content for equity markets. If related actions occur on consecutive trading days, both are retained as separate events.

6. *Verifiable source.* Every event must be traceable to at least one official government document (executive order, Federal Register notice, USTR press release, MOFCOM announcement) or, for events communicated informally, to contemporaneous reporting from at least two major wire services.

These criteria are conservative by design. Specifically, they exclude several episodes that received substantial media attention but did not constitute discrete policy actions—for example, the cumulative tariff calculations that some news outlets reported as new events when they were in fact arithmetic summations of previously announced rates.

The event set in Term 1 begins with Trump's August 14, 2017 memorandum directing the USTR to investigate China under Section 301 and includes the Section 201 safeguard tariffs on solar panels and washing machines, Section 232 steel and aluminum tariffs, the major Section 301 tariff rounds ($34B, $200B, $300B tranches), China's retaliatory tariffs, the August 2019 PBOC currency move past 7/USD, and de-escalation events including the December 2018 G20 truce and the October 2019 Phase One agreement framework. In contrast, Term 2 events begin with the February 1, 2025 executive order imposing 10% tariffs on Chinese imports under IEEPA authority and include the rapid escalation to 145% effective tariff rates by April 2025, the Geneva agreement, and subsequent adjustments and negotiations through early 2026.

Of the 44 events, 34 are classified as escalation, 9 as de-escalation, and 1 as institutional (the February 2026 Supreme Court ruling). Seven events fall on weekends (typically G20 summit announcements or social media posts); for these, the market reaction is attributed to the following Monday's trading session.

Table 3: Trade Policy Events

Date	Description	Direction	Initiator	Term
2017-08-14	Trump signs memo directing USTR to investigate China under Section 301	Escalation	US	1
2018-01-22	Trump imposes safeguard tariffs on solar panels and washing machines (Section 201)	Escalation	US	1

2018-03-08	Trump signs proclamation imposing 25% steel and 10% aluminum tariffs (Section 232)	Escalation	US	1
2018-03-22	Trump signs memo on $50B China tariffs based on Section 301 investigation	Escalation	US	1
2018-04-03	USTR publishes proposed $50B tariff list (1,333 product lines)	Escalation	US	1
2018-04-04	China announces retaliatory 25% tariffs on $50B of US goods (106 products)	Escalation	China	1
2018-06-15	Trump confirms 25% tariff on $50B Chinese goods; first $34B effective July 6	Escalation	US	1
2018-06-16	China announces retaliatory tariffs on $50B of US goods	Escalation	China	1
2018-07-06	25% tariffs on $34B take effect (both sides)	Escalation	Bilateral	1
2018-07-10	USTR releases proposed 10% tariff list on additional $200B of Chinese goods	Escalation	US	1
2018-08-01	Trump directs USTR to consider raising $200B tariff from 10% to 25%	Escalation	US	1
2018-08-03	China announces retaliatory tariffs on $60B of US goods	Escalation	China	1
2018-08-23	25% tariffs on additional $16B take effect (both sides)	Escalation	Bilateral	1
2018-09-17	Trump announces 10% tariff on $200B of Chinese goods effective Sep 24	Escalation	US	1
2018-09-24	10% tariffs on $200B (US) and $60B (China) take effect	Escalation	Bilateral	1
2018-12-01	Trump-Xi G20 dinner: 90-day truce, tariff increase delayed	De-escalation	Bilateral	1
2019-05-05	Trump tweets tariffs on $200B will increase from 10% to 25% on May 10	Escalation	US	1
2019-05-10	25% tariff on $200B of Chinese goods takes effect	Escalation	US	1
2019-05-13	China announces retaliatory tariff increase on $60B of US goods	Escalation	China	1
2019-06-29	Trump-Xi G20 meeting: truce, new tariffs suspended	De-escalation	Bilateral	1
2019-08-01	Trump announces 10% tariff on remaining $300B of Chinese imports	Escalation	US	1
2019-08-05	PBOC allows yuan to weaken past 7/USD; US labels China currency manipulator	Escalation	China	1
2019-08-13	USTR delays some tariffs on $300B list to December 15	De-escalation	US	1
2019-09-01	15% tariff on approximately $112B of Chinese goods takes effect	Escalation	US	1
2019-10-11	Phase One trade deal announced in principle	De-escalation	Bilateral	1
2019-12-13	Phase One deal text agreed; planned Dec 15 tariffs cancelled	De-escalation	Bilateral	1

2025-02-01	Trump signs EO 14195: 10% tariff on Chinese imports citing fentanyl emergency (IEEPA)	Escalation	US	2
2025-02-04	China announces retaliatory tariffs on US goods (10–15% on select products)	Escalation	China	2
2025-03-04	Additional 10% tariff on Chinese imports takes effect (cumulative 20%)	Escalation	US	2
2025-03-04	China retaliates with 10–15% tariffs on additional US agricultural products	Escalation	China	2
2025-04-02	Liberation Day: Trump announces 34% additional tariff on China (IEEPA reciprocal tariffs)	Escalation	US	2
2025-04-04	China announces 34% retaliatory tariffs on all US goods	Escalation	China	2
2025-04-08	US raises China tariff to 104% after China retaliates	Escalation	US	2
2025-04-09	Trump announces 90-day pause on reciprocal tariffs for most countries (not China)	De-escalation	US	2
2025-04-09	China tariff raised to 125%	Escalation	US	2
2025-04-10	China raises tariffs on US goods to 125%	Escalation	China	2
2025-04-12	US tariff on China effectively reaches 145% (cumulative)	Escalation	US	2
2025-05-12	US-China Geneva agreement: 90-day mutual tariff reduction (US to 30%, China reciprocal)	De-escalation	Bilateral	2
2025-08-01	US-China agree to extend Geneva truce terms	De-escalation	Bilateral	2
2025-10-10	Trump threatens 100% additional tariffs on China via Truth Social	Escalation	US	2
2025-11-10	US-China deal: tariff reduction extended for 1 year through Nov 2026	De-escalation	Bilateral	2
2026-02-20	Supreme Court strikes down IEEPA tariff regime (Learning Resources v. Trump)	Institutional	Judicial	2
2026-02-24	Trump reimposes 10% tariffs under Section 122 of Trade Act of 1974	Escalation	US	2
2026-03-11	USTR initiates new Section 301 investigations on China and others	Escalation	US	2

Note: 44 events total: 26 in Term 1 (August 2017–December 2019), 18 in Term 2 (February 2025–March 2026). Sources include White House, USTR, Federal Register, MOFCOM, and court filings. Each event is classified by direction (escalation/de-escalation), initiator (U.S./China/Bilateral/Judicial), and announcement timestamp in Eastern Time.

4.4 Supplementary Data

Daily benchmark returns are obtained from Yahoo Finance for 24 tickers spanning U.S. sectors, international indices, China-focused ETFs, and safe-haven assets; the MSCI World ex-USA ETF (ACWX) is the market benchmark for the daily event study. Two additional controls enter the HAR-RV specification. The VIX index is obtained from CBOE via Yahoo Finance, and U.S. and China Economic Policy Uncertainty indices are obtained from the EPU website maintained by Baker, Bloom, and Davis.

Daily price data from the CRSP database (accessed via WRDS) provide an alternative source for U.S. equity returns over 2016–2024. Compustat annual fundamentals and the CRSP-Compustat linking table are used to identify S&P 500 constituents and their financial characteristics.

4.5 Realized Volatility Construction

Daily RV, BV, and jump components are computed from 5-minute returns for each ETF-term combination. Days with fewer than 10 valid returns are excluded. Summary statistics appear in Table 4. Mean annualized volatility ranges from 12% to 22%, and the fraction of days with significant jumps (BNS test, 5% level) ranges from 10% to 25%—both matching existing estimates for U.S. equity ETFs.

Table 4: Realized Volatility Summary Statistics

ETF	China Exp.	Mean RV T1 ($\times 10^{-4}$)	Ann. Vol T1	Jump % T1	Mean RV T2 ($\times 10^{-4}$)	Ann. Vol T2	Jump % T2
SPY	—	0.49	11.1%	34.2%	1.08	16.5%	28.9%
XLK	0.25	0.73	13.6%	36.0%	1.63	20.3%	26.6%
QQQ	0.22	0.79	14.1%	36.9%	1.15	17.0%	27.5%
XLB	0.15	0.63	12.6%	40.1%	0.92	15.2%	44.0%
XLI	0.12	0.58	12.1%	37.5%	0.91	15.1%	39.1%
XLY	0.10	0.54	11.7%	38.8%	1.19	17.3%	30.6%

XLE	0.08	1.12	16.8%	33.7%	1.52	19.6%	40.3%
XLV	0.07	0.52	11.4%	38.4%	0.74	13.7%	45.1%
XLC	0.06	0.92	15.2%	40.6%	0.86	14.7%	35.0%
XLF	0.03	0.78	14.0%	32.6%	1.00	15.9%	41.9%

Note: RV = realized variance (sum of squared 5-minute returns). BV = bipower variation (Barndorff-Nielsen and Shephard, 2004). Jump = max(RV − BV, 0). BNS Jump % = fraction of days with significant jumps at the 5% level (Barndorff-Nielsen and Shephard, 2006). Annualized volatility = sqrt(252 × mean daily RV). Days with fewer than 10 valid returns are excluded.

5. Results

5.1 U.S. Daily Event Study

Table 5 reports the daily event study results for key U.S. ETFs using both the standard BMP test and the Bremer and Zhang (2007) GARCH-based extension. The standard BMP test fails to reject the null of no abnormal returns for most ETF-window-term combinations. In contrast, the Bremer-Zhang test, by accounting for event-induced variance, reaches significance in several cases where the standard test does not, though for XLI in Term 1 both tests are significant. For SPY in Term 2, the CAR[-1,+1] window yields $t_{BZ} = -2.12$ ($p = 0.049$), significant at the 5% level, whereas the standard BMP statistic is $t_{BMP} = -1.05$ ($p = 0.307$). For XLF in Term 2, the event-day abnormal return is significant under the GARCH-adjusted test ($t_{BZ} = -2.14$, $p = 0.047$), and the CAR[-1,+1] window is also significant ($t_{BZ} = -2.24$, $p = 0.038$). These results highlight the importance of accounting for GARCH effects in event studies of trade policy shocks, where event-day variance is unlikely to be constant across events.

Table 5: Daily Event Study — BMP vs. Bremer-Zhang (2007) Comparison

Ticker	Term	Window	t_{BMP}	p_{BMP}	t_{BZ}	p_{BZ}	N
SPY	1	AR[0]	0.84	0.407	0.14	0.887	25

SPY	2	AR[0]	0.02	0.983	-0.83	0.417	18
SPY	2	CAR[-1,+1]	-1.05	0.307	-2.12	0.049**	18
SPY	2	CAR[-2,+2]	-0.57	0.579	-2.00	0.062*	18
XLF	1	AR[0]	-0.32	0.750	-0.09	0.932	25
XLF	2	AR[0]	-1.20	0.245	-2.14	0.047**	18
XLF	2	CAR[-1,+1]	-1.83	0.086	-2.24	0.038**	18
XLI	1	CAR[-2,+2]	-2.81	0.009	-2.70	0.012**	26
XLI	2	CAR[-1,+1]	-1.13	0.276	-1.99	0.063*	18

*Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. BMP: Boehmer, Musumeci, and Poulsen (1991). BZ: Bremer and Zhang (2007) GARCH-based extension. The table shows AR[0] for all ETFs and additional windows where significance is detected.*

5.2 Chinese Daily Event Study

Table 6 reports the Chinese daily event study results. In contrast to the U.S. results, Chinese markets show a declining pattern of abnormal returns from Term 1 to Term 2. The Consumer ETF (159928) is the most consistently significant, with CAR[-2,+2] in Term 2 yielding $t_{BMP} = 3.28$ ($p = 0.007$) and $t_{BZ} = 3.77$ ($p = 0.003$). The CSI 300 ETF shows no significant effects in either term under either test. The declining Chinese reaction is confirmed by the cross-term comparison: for the CSI 300 ETF, mean $|AR[0]|$ falls from 1.61% in Term 1 to 1.13% in Term 2, though this difference is not statistically significant (Wilcoxon $p = 0.081$; permutation $p = 0.436$).

Table 6: Chinese Daily Event Study — BMP and GARCH-adjusted Tests

ETF	Term	Window	t_{BMP}	p_{BMP}	t_{BZ}	p_{BZ}	N
CSI300	1	AR[0]	-0.85	0.403	-0.82	0.420	26
CSI300	2	AR[0]	-0.42	0.681	-0.35	0.736	12
SSE50	1	AR[0]	-0.48	0.636	-0.38	0.709	26
SSE50	2	AR[0]	-0.60	0.560	-0.54	0.601	12
SSE50	2	CAR[-2,+2]	1.29	0.225	1.96	0.076*	12
GEI	1	AR[0]	-0.57	0.571	-0.10	0.922	26
GEI	2	AR[0]	-0.59	0.568	-0.62	0.549	12
Consumer	1	AR[0]	-0.74	0.469	-0.36	0.720	26

Consumer	2	AR[0]	-1.15	0.273	-1.63	0.131	12
Consumer	2	CAR[-2,+2]	3.28	0.007***	3.77	0.003***	12
Securities	1	AR[0]	-1.17	0.252	-1.07	0.295	26
Securities	2	AR[0]	-0.32	0.753	-0.15	0.884	11
Bank	1	AR[0]	0.03	0.980	0.03	0.977	25
Bank	2	AR[0]	-0.50	0.624	-0.37	0.721	12
Medical	1	AR[0]	-0.28	0.784	-0.02	0.984	26
Medical	2	AR[0]	-0.48	0.641	-0.14	0.890	12

Significance levels: $^{***}p < 0.01$*,* $^{**}p < 0.05$*,* $^{*}p < 0.10$*. The table shows AR[0] for all ETFs and additional windows where significance is detected. Bank ETF (512800) has N = 25 in Term 1 because one event predates its listing.*

5.3 U.S. Intraday Event Study

Table 7 reports the intraday event study results for SPY. The difference between terms is visible even at the single-bar level: the immediate event-bar reaction is 2.4 times larger in Term 2 than in Term 1 (|AR[0]| = 0.162% vs. 0.068%, permutation $p = 0.004$). The gap widens at longer horizons. In the ±60-minute window, mean |CAR| is 1.152% in Term 2 versus 0.389% in Term 1, approximately three times larger, and the difference remains significant at the 1% level ($p = 0.004$). The full-day window yields a difference of 0.869 percentage points at borderline significance ($p \approx 0.05$ across multiple permutation runs), and the weaker significance likely reflects the accumulation of non-event noise over the longer horizon.

The cross-term magnitude comparison reveals a consistent pattern: Term 2 mean |CAR| exceeds Term 1 across every intraday window, with permutation tests rejecting the null of equal magnitudes at the 1% level for the $\pm$60-minute window (Term 2 |CAR| = 1.152% vs. Term 1 |CAR| = 0.389%, $p = 0.004$). Nonetheless, the BMP cross-sectional tests on Term 2 alone are insignificant for most windows—the strongest is CAR[0,+12] at $t_{BMP} = -1.81$ ($p = 0.088$)—because second-term events drove large price moves in both directions rather than systematically one way. In contrast, Term 1 BMP statistics are generally insignificant, with the notable exception of the pre-event window ($t_{BMP} = 2.67$, $p = 0.014$), consistent with anticipatory positioning ahead of announcements that were frequently previewed through official channels during the first term.

The pre-event abnormal returns in Term 2 are also noteworthy. The CAR[-12,-1] window shows mean |CAR| of 0.901% versus 0.242% in Term 1 ($p = 0.008$), which suggests more aggressive pre-announcement positioning by market participants, in line with publicly visible policy signals via social media and press conferences that characterized the second term's announcement style. Understandably, participants who had learned from the first term to trade on early signals may have front-run formal announcements more aggressively.

Table 7: Intraday Event Study - Cross-Term Comparison (SPY)

Window	T1 Mean |CAR|	T2 Mean |CAR|	Ratio	Rank-sum p	Perm p
AR[0]	0.068%	0.162%	2.368	0.010**	0.004***
CAR[-6,0]	0.200%	0.517%	2.585	0.006***	0.024**
CAR[-12,0]	0.260%	0.973%	3.742	0.020**	0.002***
CAR[0,+6]	0.204%	0.313%	1.534	0.028**	0.104
CAR[0,+12]	0.267%	0.552%	2.070	0.008***	0.016**
CAR[-6,+6]	0.326%	0.517%	1.585	0.150	0.218
CAR[-12,+12]	0.389%	1.152%	2.963	0.018**	0.004***
CAR[0,+78]	0.675%	1.544%	2.288	0.047**	0.049*
CAR[-12,-1]	0.242%	0.901%	3.720	0.004***	0.007***

*Note: T1 N = 25 events, T2 N = 18. |AR[0]| and |CAR| are mean absolute abnormal returns across events within each term. Permutation test with 10,000 replications of term labels. *** p < 0.01, ** p < 0.05, * p < 0.10.*

Table 8 reports the analogous Chinese intraday comparison for the CSI 300 ETF. Conversely, the pattern is the opposite of the U.S. results. In the opening windows (15 minutes, 30 minutes, first hour), Term 2 reactions are modestly larger than Term 1 (ratios of 1.27–1.39), but none of these differences reach statistical significance. Over longer horizons, the relationship reverses: the morning session ratio falls to 0.79 and the full-day ratio to 0.88, in line with the daily event study finding that Chinese reactions attenuated in Term 2. The last-hour window shows a statistically significant decline (Term 1: 0.326%, Term 2: 0.116%, ratio 0.36, permutation $p = 0.031$). This pattern suggests that Chinese markets priced in trade events more quickly in Term 2 and that the initial intraday reaction dissipated by session end. The asymmetry between U.S. and Chinese intraday reactions — amplification versus attenuation — is the intraday-frequency analog of the cross-market DID result in Section 5.4.

Table 8: Chinese Intraday Event Study - Cross-Term Comparison (CSI 300 ETF)

Window	T1 Mean	CAR		T2 Mean
Opening 15 min	0.324%	0.450%	1.389	0.378
Opening 30 min	0.411%	0.531%	1.291	0.519
First hour	0.482%	0.612%	1.270	0.459
Morning session	0.813%	0.639%	0.785	0.577
Full day	0.957%	0.844%	0.882	0.748
Last hour	0.326%	0.116%	0.356	0.031**

Note: T1 N = 26 events (10 for Last hour), T2 N = 11 (one Term 2 event date falls on a Chinese market holiday). Permutation test with 10,000 replications. The Last hour window uses only events occurring before the afternoon session close.

5.4 Cross-Market Difference-in-Differences

The intraday event study establishes that U.S. reactions amplified in Term 2. To test whether U.S. and Chinese markets diverged in their response trajectories, I estimate a cross-market DID using absolute cumulative abnormal returns. The estimand is a difference-in-magnitudes rather than a standard treatment-effect DID, since the outcome variable $|CAR|$ is non-negative by construction. I note this distinction because the usual parallel-trends interpretation does not carry over directly; the DID here measures whether the gap in reaction magnitudes between U.S. and Chinese markets widened from Term 1 to Term 2.

The permutation-based DID at the immediate event-day window (AR[0]) is 1.29 percentage points ($p = 0.040$). This combines two effects: U.S. reactions amplified while Chinese reactions attenuated. The result survives leave-one-out exclusion of any single event and holds across the basic regression specification with clustered standard errors ($\hat{\beta}_3 = 0.019$, $p = 0.003$).

However, at wider windows, the pattern attenuates. With event fixed effects, which absorb event-specific heterogeneity and provide a more demanding test, the interaction coefficient at CAR[-1,+1] has $p = 0.419$ and at CAR[-2,+2] has $p = 0.318$. The honest interpretation: the cross-market divergence is sharpest at the immediate announcement reaction and becomes noisier at longer horizons as non-event-day variation accumulates. The event-FE specification is more credible but also more demanding of a sample with only 12 paired Chinese event observations in Term 2.

As a further check, I run a placebo test that randomly permutes term labels across events 10,000 times and recomputes the DID under each permutation. The actual AR[0] DID falls at the 99.9th percentile of the single-market placebo distribution (permutation p = 0.002) and the 99th percentile of the cross-market distribution (p = 0.010). The wider-window results are weaker but directionally consistent. Appendix Table 15 reports the full placebo results. Separately, Appendix Table 14 reports mean signed CARs by event direction and term, confirming that the absolute-value DID reflects genuine directional asymmetry: Term 2 escalation events produced larger negative returns and Term 2 de-escalation events produced larger positive returns relative to Term 1.

Table 9: Cross-Market DID

Window	Permutation DID	Perm p	Regression $\hat{\beta}_3$	Basic p	Event FE p
AR[0]	0.0129	0.040**	0.0126	0.094*	0.0131

CAR[-1,+1]	0.0071	0.341	0.0057	0.527	0.0079
CAR[-2,+2]	0.0062	0.397	0.0054	0.550	0.0106

*Note: DID = (|CAR|_US,T2 − |CAR|_US,T1) − (|CAR|_CN,T2 − |CAR|_CN,T1), reported in percentage points. U.S. = SPY, China = CSI 300 ETF. "Perm. p" is the two-sided p-value from 10,000 permutations of term labels within each market. *** p < 0.01, ** p < 0.05, * p < 0.10.*

5.5 HAR-RV Model Results

With only 18 second-term events against substantial daily noise in realized volatility, the HAR-RV interaction tests are underpowered; I treat them as suggestive evidence on a distinct quantity. The intraday event study measures directional price moves (first-moment: how far prices moved). The HAR-RV model measures volatility dispersion (second-moment: how much prices bounced around within the day). These can diverge. If trade events cause sharp one-directional repricing rather than intraday whipsawing, CARs will amplify while RV interactions remain small.

Table 10 reports the HAR-RV estimation results. The baseline model fits reasonably well, with R^2 ranging from 0.41 to 0.63 across tickers, which is in line with what Corsi (2009) and subsequent studies have found for U.S. equities. The event dummy ($\hat{\delta}_1$) is positive for all tickers and reaches marginal significance for SPY ($\hat{\delta}_1 = 0.25$, $t = 1.66$), QQQ ($\hat{\delta}_1 = 0.54$, $t = 1.72$), and XLI ($\hat{\delta}_1 = 0.24$, $t = 1.83$). In economic terms, these estimates imply that trade policy event days are associated with 25–50% higher log realized volatility after controlling for persistence, which is substantial.

The event×Term 2 interaction coefficient ($\hat{\delta}_3$) is positive for 8 of 10 sector ETFs, with point estimates ranging from 0.09 to 0.38, but no individual estimate reaches the 10%

significance level. Indeed, the consistent sign pattern across sectors is suggestive of a real effect, but the lack of statistical significance is not surprising given that there are only 18 second-term events in the sample, working against substantial daily noise in realized volatility. Adding the log VIX as a control improves R^2 by 6–10 percentage points and reveals a notable pattern: the Term 2 level dummy ($\hat{\delta}_2$) turns significantly negative, in the range of -0.2 to -0.3. This implies that non-event-day volatility is lower in the second term, conditional on VIX. The second term is thus characterized by compressed baseline volatility punctuated by sharp spikes on event days; this pattern is distinct from the first term's more diffuse volatility profile.

Table 10: U.S. HAR-RV Model - Event-Term Interaction Coefficients

Ticker	Event Coef	Event SE	Event×Term2 Coef	Event×Term2 SE	N	R²
SPY	0.2580*	(0.1541)	0.1751	(0.3277)	1284	0.614
QQQ	0.5413*	(0.3116)	-0.1239	(0.3969)	1284	0.570
XLF	0.1428	(0.1582)	0.3786	(0.2535)	1283	0.489
XLI	0.2436*	(0.1302)	0.2473	(0.2432)	1283	0.606
XLK	0.1933	(0.1630)	0.1769	(0.3151)	1283	0.637
XLB	0.1204	(0.1156)	0.3695	(0.2340)	1284	0.548
XLE	0.2550	(0.1905)	0.0927	(0.2766)	1284	0.450
XLV	0.2035*	(0.1209)	0.2713	(0.2118)	1283	0.588
XLC	0.2652	(0.1883)	0.1754	(0.2754)	790	0.499
XLY	0.2558*	(0.1552)	0.1271	(0.2660)	1283	0.649

Significance levels: $***\, p < 0.01$*,* $**\, p < 0.05$*,* $*\, p < 0.10$. *XLP and XLU are excluded because their Term 2 data are unavailable, precluding estimation of the interaction term.*

For Chinese ETFs, the HAR-RV results tell a complementary story. The event dummy ($\hat{\delta}_1$) is large and highly significant for the CSI 300 ETF ($\hat{\delta}_1 = 0.52$, $t = 3.71$, $p < 0.001$), indicating that Chinese markets do react to trade policy events on average. Crucially, the event×Term 2 interaction ($\hat{\delta}_3$) is negative for all seven Chinese ETFs—the opposite sign from the U.S. results, though none reach statistical significance (CSI 300: $\hat{\delta}_3 = -0.24$, $t = -0.56$; SSE 50: $\hat{\delta}_3 = -0.16$, $t = -0.34$; Securities: $\hat{\delta}_3 = -0.32$, $t = -0.91$). The Chinese HAR-RV results corroborate the daily event study finding that Chinese markets reacted less in Term 2 than in Term 1.

Table 11: Chinese HAR-RV Model - Event-Term Interaction Coefficients

ETF	Event Coef	Event SE	Event×Term2 Coef	Event×Term2 SE	N	R^2
CSI300	0.5203***	(0.1402)	-0.2449	(0.4377)	1063	0.500
SSE50	0.3947***	(0.1253)	-0.1630	(0.4743)	1065	0.530
GEI	0.3838***	(0.1216)	-0.0280	(0.4240)	1063	0.528
Medical	0.4013	(0.3095)	-0.2162	(0.3748)	620	0.302
Securities	0.3919**	(0.1565)	-0.3170	(0.3488)	862	0.370
Bank	0.1011	(0.1788)	-0.2062	(0.3770)	515	0.311
Consumer	0.2219	(0.2976)	-0.0710	(0.4747)	416	0.477

Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$

5.6 Sectoral Analysis

Detectable effect size. Before examining the regression results, a power calculation sets expectations. The standard error on the triple interaction$\hat{\delta}_7$ is 1.13, yielding a minimum detectable effect at 80% power of 3.17 log RV points. The ex-post power for the observed coefficient is 10.2%. Any sector-specific amplification smaller than roughly three times the standard error would go undetected in this sample.

With that caveat, Table 12 reports the sectoral panel results for U.S. ETFs. China revenue exposure across the 10 sectors ranges from 25% for technology (XLK) to 1% for utilities (XLU), constructed from Compustat geographic segment data following the methodology of Huang, Lin, Liu, and Tang (2023). Despite this wide range, there is no significant relationship between exposure and the magnitude of intraday reactions in either term. In Term 1, the high-exposure group actually has slightly lower mean |CAR| than the low-exposure group (0.547% vs. 0.741%, $p = 0.996$). The triple interaction coefficient is $\hat{\delta}_7 = -0.78$ ($t = -0.69, p = 0.49$). Given the low power, I cannot reject the null of no sector-specific amplification, but neither can I detect plausible heterogeneity. What the data do show is that any such heterogeneity is not first-order relative to the broad market effect.

Two interpretations remain open. The sector-level exposure estimates may be too coarse; firm-level Compustat geographic segments would provide a sharper test. Conversely, the null may be real: the 2025 escalation involved reciprocal tariffs on virtually all trading partners with considerable uncertainty about exemptions and phase-ins, affecting market-wide risk perceptions rather than sector-specific expected cash flows.

Table 12: Panel Regression - Sectoral Analysis

Panel A: Pooled Model

Variable	Coefficient	Std Error	t-stat	p-value
const	-0.3007***	(0.0595)	-5.06	0.0000
log_rv_lag	0.9702***	(0.0059)	165.65	0.0000
event	0.1523**	(0.0592)	2.57	0.0101
china_exposure	-0.0021	(0.0531)	-0.04	0.9679
event_x_exposure	0.3294	(0.5501)	0.60	0.5494

Panel B: Term Interactions Model

Variable	Coefficient	Std Error	t-stat	p-value
const	-0.3442***	(0.0622)	-5.54	0.0000
log_rv_lag	0.9660***	(0.0061)	158.10	0.0000
event	0.0870	(0.0693)	1.26	0.2090
china_exposure	-0.0300	(0.0623)	-0.48	0.6295
term2	0.0066	(0.0151)	0.43	0.6645
event_x_exposure	0.5042	(0.6954)	0.73	0.4684
event_x_term2	0.2253*	(0.1247)	1.81	0.0709
exposure_x_term2	0.0720	(0.1236)	0.58	0.5599
triple	-0.7771	(1.1296)	-0.69	0.4915

Significance levels: *** p<0.01, ** p<0.05, * p<0.10. N = 13,038 observations.

5.7 Escalation versus De-escalation: Directional Evidence

The magnitude DID in Section 5.4 uses absolute CARs and cannot distinguish between directional channels. Disaggregating events by direction provides cleaner identification because the directional predictions differ across mechanisms. If amplification operates through expected cash flows, escalation events should produce larger negative returns and de-escalation events should produce larger positive returns. Under the stochastic discount factor channel (Pastor and Veronesi, 2012), both directions should produce larger reactions, because any policy change raises uncertainty regardless of direction.

The data favor the discount-factor channel. Escalation events produce mean signed CARs of $+0.178\%$ in Term 2 compared to -0.035% in Term 1 in the ±30-minute window, with absolute values roughly five times larger. De-escalation events, however, remain positive in both terms but attenuate sharply: the mean signed CAR falls from $+0.712\%$ in Term 1 to $+0.196\%$ in Term 2. Two of four Term 2 de-escalation events produce negative CARs (Geneva agreement: -0.52%; truce extension: -0.46%), while the April 9, 2025 pause generates a large positive reaction ($+1.58\%$) despite the simultaneous increase in China-specific rates to 125%. The attenuation of de-escalation reactions suggests diminished credibility of concessions in Term 2.

In the HAR-RV specification, de-escalation events show positive and often significant volatility coefficients. Both escalation and de-escalation events elevate realized volatility. This symmetry is the prediction of Pastor and Veronesi (2012): policy changes command a risk premium because they increase the variance of the policy-state variable, and this effect does not depend on the direction of the change. The implication for the two-term comparison is that the amplification reflects an increase in perceived policy risk rather than a deterioration in expected trade outcomes per se. Even conciliatory gestures in Term 2 carried more uncertainty than escalatory moves in Term 1.

5.8 Macro Announcement Overlap

Because trade policy events occasionally coincide with scheduled macroeconomic releases, I cross-reference the 44 event dates against FOMC announcement dates and nonfarm payroll (NFP) release dates. One event directly overlaps with an FOMC announcement day: August 1, 2018 (Trump directs USTR to consider raising the $200B tariff, same day as the FOMC statement). Four events fall on NFP release dates: July 6, 2018; August 3, 2018; April 4, 2025; and August 1, 2025. Two additional events fall within one calendar day of an FOMC decision: March 22, 2018 is one day after the March 21 FOMC, and August 1, 2019 is one day after the July 31 FOMC. In total, 5 of 44 events (11.4%) have same-day overlap with a major macro release. Re-estimating the cross-market DID after excluding these 5 events yields a stronger result: the AR[0] DID coefficient rises from 1.29 percentage points (permutation $p = 0.040$) in the full sample to 1.86 percentage points (permutation $p = 0.001$) in the restricted sample. The OLS regression confirms the pattern, with the interaction term significant at the 0.1% level ($t = 3.58$, $p < 0.001$). Wider windows follow the same direction but remain statistically insignificant, consistent with their baseline estimates. The fact that excluding macro-coincident events strengthens rather than weakens the DID indicates that FOMC and NFP releases were injecting noise that diluted the tariff-specific signal. This reinforces the identifying assumption that the measured asymmetry reflects trade policy risk, not generalized macroeconomic news (full results in Appendix Table 13).

6. Discussion

The findings support a learning interpretation of market reactions to trade policy shocks, though not the one that standard Bayesian updating models would predict. In a simple learning framework, repeated exposure to trade policy shocks should reduce surprise over time and hence reduce volatility—investors learn the distribution of possible outcomes and update their priors accordingly. In fact, I find the opposite: markets reacted more in the second term, not less. Sequencing explains the paradox. Gradual escalation in the first term, followed by resolution through the Phase One agreement, likely tightened investor posteriors around a range of moderate outcomes: tariff rates of 10–25%, temporary disruptions, eventual negotiated settlement. The second term's roughly 145% effective tariff rates on Chinese imports and broad reciprocal tariffs on many trading partners fell well outside the support of those posteriors. In Bayesian terms, the first term's experience made the second term's realizations *more* surprising, not less, because it narrowed the set of outcomes investors considered plausible.

The fact that the amplification is broad rather than sector-specific is also informative. If the mechanism were primarily about firm-level trade exposure (Chinese supply chains, revenue dependence on the Chinese market), one would expect sectors like technology and materials to react more than utilities and financials. This is not, however, what I find. The null result on the triple interaction ($p = 0.49$) is more consistent with models in which policy uncertainty affects the stochastic discount factor directly, as in Pastor and Veronesi (2012, 2013), rather than operating through sector-specific cash flow channels.

Macro Environment and Market Structure Differences Between Terms: The two-term comparison is not a randomized experiment. The macro environment differed between terms in ways that could independently affect market reactions. The federal funds rate stood near 2% during most of Term 1 and above 4% during Term 2. Equity valuations, particularly in technology, were elevated in 2025 relative to 2018, partly reflecting AI-related speculation. The post-COVID inflation regime and associated Fed tightening cycle had no parallel in the first term. U.S. equity market microstructure also changed: the explosion of zero-days-to-expiration (0DTE) options after 2022 altered intraday SPY dynamics through gamma-hedging flows, potentially amplifying intraday moves independently of trade policy. These differences are real confounders. The identification strategy partially addresses them: intraday abnormal returns are computed relative to a 60-day rolling estimation window, which nets out much of the low-frequency drift in baseline volatility and valuations. Event fixed effects in the regression DID absorb event-specific macro states. The finding that the amplification is strongest at the narrowest window (the immediate event bar) and weakest at full-day horizons points to a trade-specific information shock rather than a general high-volatility environment. The VIX control in the HAR-RV specification absorbs market-wide fear. I cannot fully rule out residual confounding from the changed macro and microstructure environment, but the pattern of results is more consistent with trade-specific amplification than with a general volatility-regime shift.

Alternative Explanations for Chinese Market Attenuation: The cross-market DID attributes the declining Chinese reaction to investor learning, but at least three alternative mechanisms deserve consideration. First, China's "National Team" (国家队), a consortium of state-backed funds, is known to intervene in A-share markets during periods of stress, purchasing index constituents to stabilize prices. If state buying intensified in Term 2 relative to Term 1, the attenuation could reflect government stabilization rather than organic learning by private investors. Without transaction-level data identifying state fund activity, I cannot separate these channels. Second, A-share circuit breakers and individual stock 10% daily price limits compress observable volatility mechanically. If more stocks approached these limits in Term 2, measured ETF-level reactions would appear smaller even absent any change in underlying fundamentals. Third, the onshore-offshore wedge complicates interpretation. A-share trading hours do not overlap with U.S. event timing; by the time Shanghai opens, H-shares in Hong Kong and offshore CNH have already priced the news. The finding that Chinese ETFs incorporate most of the reaction in the first 15 minutes of trading may reflect overnight price discovery in offshore venues rather than learning by onshore investors. Adding Hong Kong-listed H-share ETFs as a third market would help distinguish these channels, since Hong Kong is more open to foreign flows and less subject to state intervention than the A-share market.

Limitations: The most consequential data gap is the absence of Hong Kong-listed H-share ETFs. Because A-share trading hours do not overlap with U.S. event timing, the Chinese reaction I measure reflects the opening response after an overnight information gap. Hong Kong H-shares trade during hours that partially overlap with U.S. afternoons, and they are accessible to foreign institutional investors without QFII quotas. Adding HSCEI-linked ETFs as a third market would allow a direct test of whether the Chinese attenuation reflects onshore A-share dynamics (National Team intervention, circuit breakers) or a broader pattern that extends to the more internationally integrated Hong Kong market. Retrospectively, this extension is the most natural next step for the cross-market analysis.

Generally speaking, the small number of Term 2 events (18 as of March 2026) limits statistical power for the HAR-RV interaction tests, and a longer sample may yield significant estimates where the current point estimates are suggestive but noisy. The Chinese intraday data from RESSET are at 5-minute resolution rather than tick-level, which precludes the microstructure analysis possible with TAQ data. The sector-level exposure estimates, constructed from revenue data at the ETF level, are coarse; firm-level data from Compustat geographic segments would provide a sharper test. The event set is manually curated from official government sources, and some policy actions communicated through social media or leaked briefings may be missing from the sample.

If trade policy uncertainty elevates equity risk premia—and the evidence from Pastor and Veronesi (2012) and Brogaard and Detzel (2015) suggests that it does—then the second term's trade policy posture imposes larger capital market costs than the first term's more gradual escalation. Hence, the dynamic costs of policy unpredictability appear to compound with repeated use rather than attenuate, which has implications for how policymakers weigh the costs and benefits of aggressive trade strategies.

7. Conclusion

This paper yields four principal findings. First, U.S. intraday abnormal returns around U.S.–China trade policy announcements are approximately two to three times larger in Trump's second presidential term than in his first; the amplification holds across event windows, test statistics, and sectors. Second, Chinese market reactions attenuated over the same period, and the cross-market DID confirms that U.S. and Chinese markets moved in opposite directions: a 1.29 percentage point divergence at the event-day level (permutation $p = 0.040$). Third, markets did not habituate to trade policy shocks. By several measures, U.S. markets were more surprised in the second term, while Chinese markets adapted. Fourth, the evidence is consistent with differential belief revision: the first term's resolution through the Phase One agreement narrowed U.S. investor posteriors, making the second term's escalation a tail realization; Chinese investors, with longer exposure to trade frictions, had already incorporated extreme outcomes into their priors.

Several extensions would strengthen and generalize these findings. A formal placebo test would draw pseudo-event dates at random from non-event trading days, matched by weekday and term, and re-run the full intraday event study on these placebo samples. If the cross-term amplification is specific to trade policy announcements rather than an artifact of generally higher Term 2 volatility, the placebo distribution of cross-term ratios should be centered near 1.0, and the actual ratios should fall in the extreme tail. This test would provide a sharp null against which to benchmark the main results and is straightforward to implement with the existing permutation infrastructure.

Adding Hong Kong-listed H-share ETFs as a third market would help separate the National Team stabilization, circuit breaker, and offshore-pricing channels from genuine learning effects in the Chinese attenuation result. Firm-level geographic segment data from Compustat would provide a sharper test of the exposure channel than the sector-level estimates used here. The options market, through implied volatility surfaces and the volatility risk premium around trade policy events, offers a complementary window into how policy uncertainty is priced, one that could help distinguish between the broad uncertainty and sector-specific channels that the current data leave unresolved.

Data Availability

TAQ and RESSET data are available via subscription from NYSE and RESSET Financial Research Database; licensing restrictions prevent public redistribution of raw trade records. The full event list, analysis code, and derived series (daily realized volatility, cumulative abnormal returns) are available at https://github.com/danxudanxu/learning-to-price-political-risk. Researchers with TAQ and RESSET access can reproduce all figures and tables exactly.

References

Amiti, M., Redding, S. J., and Weinstein, D. E. (2019). The impact of the 2018 tariffs on prices and welfare. *Journal of Economic Perspectives*, 33(4), 187–210.

Andersen, T. G., Bollerslev, T., Diebold, F. X., and Labys, P. (2001). The distribution of realized exchange rate volatility. *Journal of the American Statistical Association*, 96(453), 42–55.

Andersen, T. G., Bollerslev, T., Diebold, F. X., and Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2), 579–625.

Andrews, D. W. K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3), 817–858.

Baker, S. R., Bloom, N., and Davis, S. J. (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4), 1593–1636.

Barndorff-Nielsen, O. E. and Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B*, 64(2), 253–280.

Barndorff-Nielsen, O. E. and Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1), 1–37.

Barndorff-Nielsen, O. E. and Shephard, N. (2006). Econometrics of testing for jumps in financial economics using bipower variation. *Journal of Financial Econometrics*, 4(1), 1–30.

Beber, A. and Brandt, M. W. (2009). Resolving macroeconomic uncertainty in stock and bond markets. *Review of Finance*, 13(1), 1–45.

Boehmer, E., Musumeci, J., and Poulsen, A. B. (1991). Event-study methodology under conditions of event-induced variance. *Journal of Financial Economics*, 30(2), 253–272.

Bollerslev, T., Patton, A. J., and Quaedvlieg, R. (2016). Exploiting the errors: A simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1), 1–18.

Bremer, R. and Zhang, Z. (2007). Extensions of the standardized cross-sectional approach to short-horizon event studies. *Journal of Financial Research*, 30(4), 495–513.

Brogaard, J. and Detzel, A. (2015). The asset-pricing implications of government economic policy uncertainty. *Management Science*, 61(1), 3–18.

Brunnermeier, M. K., Sockin, M., and Xiong, W. (2022). China's model of managing the financial system. *Review of Economic Studies*, 89(6), 3115–3153.

Caldara, D., Iacoviello, M., Molligo, P., Prestipino, A., and Raffo, A. (2020). The economic effects of trade policy uncertainty. *Journal of Monetary Economics*, 109, 38–59.

Cheng, M., Guo, L., Hu, R., and Zheng, H. (2022). The effectiveness of government stock purchase during market crash: Evidence from China. *Pacific-Basin Finance Journal*, 71, 101679.

Collin-Dufresne, P., Johannes, M., and Lochstoer, L. A. (2016). Parameter learning in general equilibrium: The asset pricing implications. *American Economic Review*, 106(3), 664–698.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.

Dang, T. V., Gorton, G., Holmstrom, B., and Ordonez, G. (2023). Government stock purchase undermines price informativeness: Evidence from China's "National Team." *Journal of Financial and Quantitative Analysis*, 58(6), 2451–2482.

Ederington, L. H. and Lee, J. H. (1993). How markets process information: News releases and volatility. *Journal of Finance*, 48(4), 1161–1191.

Egger, P. H. and Zhu, J. (2020). The US–Chinese trade war: An event study of stock-market responses. *Economic Policy*, 35(103), 519–559.

Fajgelbaum, P. D., Goldberg, P. K., Kennedy, P. J., and Khandelwal, A. K. (2020). The return to protectionism. *Quarterly Journal of Economics*, 135(1), 1–55.

Greenland, A., Ion, M., Lopresti, J., and Schott, P. K. (2020). Using equity market reactions to infer exposure to trade liberalization. *NBER Working Paper No. 27510*.

Handley, K. and Limao, N. (2017). Policy uncertainty, trade, and welfare: Theory and evidence for China and the United States. *American Economic Review*, 107(9), 2731–2783.

Handley, K. and Limao, N. (2022). Trade policy uncertainty. *Annual Review of Economics*, 14, 363–395.

Huang, Y., Lin, C., Liu, S., and Tang, H. (2023). Trade networks and firm value: Evidence from the US–China trade war. *Journal of International Economics*, 145, 103811.

Kelly, B., Pastor, L., and Veronesi, P. (2016). The price of political uncertainty: Theory and evidence from the option market. *Journal of Finance*, 71(5), 2417–2480.

Kozlowski, J., Veldkamp, L., and Venkateswaran, V. (2020). The tail that wags the economy: Beliefs and persistent stagnation. *Journal of Political Economy*, 128(8), 2839–2879.

Li, H., Zhang, X., and Zhao, R. (2024). China's National Team: A game changer in stock market stabilization? *Finance Research Letters*, 60, 104887.

Malmendier, U. and Nagel, S. (2011). Depression babies: Do macroeconomic experiences affect risk taking? *Quarterly Journal of Economics*, 126(1), 373–416.

Pastor, L. and Veronesi, P. (2012). Uncertainty about government policy and stock prices. *Journal of Finance*, 67(4), 1219–1264.

Pastor, L. and Veronesi, P. (2013). Political uncertainty and risk premia. *Journal of Financial Economics*, 110(3), 520–545.

Patton, A. J. and Sheppard, K. (2015). Good volatility, bad volatility: Signed jumps and the persistence of volatility. *Review of Economics and Statistics*, 97(3), 683–697.

Wagner, A. F., Zeckhauser, R. J., and Ziegler, A. (2018). Company stock price reactions to the 2016 election shock: Trump, taxes, and trade. *Journal of Financial Economics*, 130(2), 428–451.

Appendix

Table 13: Cross-Market DID Robustness — Excluding FOMC/NFP Overlap Events

	Full Sample (N = 44)			**Excl. FOMC/NFP (N = 39)**		
Window	**DID (pp)**	**Perm. p**	**Reg. p**	**DID (pp)**	**Perm. p**	**Reg. p**
AR[0]	1.29	0.040**	0.068*	1.86	0.001***	0.001***
CAR[−1,+1]	0.71	0.341	0.506	1.11	0.146	0.318
CAR[−2,+2]	0.62	0.397	0.621	0.80	0.296	0.553

*Notes: DID = (|CAR|_US,T2 − |CAR|_US,T1) − (|CAR|_CN,T2 − |CAR|_CN,T1), reported in percentage points. "Perm. p" is the two-sided p-value from 10,000 permutations of term labels within each market. "Reg. p" is from OLS with HC1 robust standard errors. The excluded events are: July 6, 2018 (NFP); August 1, 2018 (FOMC); August 3, 2018 (NFP); April 4, 2025 (NFP); August 1, 2025 (NFP). *** p < 0.01, ** p < 0.05, * p < 0.10.*

Table 14: Mean Signed Cumulative Abnormal Returns by Direction and Term (SPY)

Window	**Escalation**		**De-escalation**	
	Term 1	**Term 2**	**Term 1**	**Term 2**
AR[0]	+0.107	−0.209	−0.087	+1.397
(SE)	(0.100)	(0.458)	(0.212)	(1.213)
CAR[-1,+1]	+0.296	−0.498	−0.212	+0.668
(SE)	(0.234)	(0.691)	(0.248)	(0.690)
CAR[-2,+2]	+0.124	−0.780	−0.280	+1.163
(SE)	(0.236)	(0.627)	(0.341)	(1.314)
N events	21	13	5	4

Notes: Mean signed CAR for SPY (S&P 500 ETF) by event direction and presidential term, in percentage points. Standard errors of the mean in parentheses. Escalation events include tariff announcements, investigation launches, and retaliatory measures. De-escalation events include trade truces, exemptions, and negotiation progress. One Term 2 event classified as "Institutional" (Supreme Court ruling) is excluded. The sign pattern confirms that the absolute-value DID in Table 9 reflects genuine directional asymmetry: Term 2 escalation events produced larger negative returns, while Term 2 de-escalation events produced larger positive returns, compared to their Term 1 counterparts.

Table 15: Placebo Test — Permutation of Term Labels

Window	SPY Only			Cross-Market (US vs CN)		
	DID (pp)	Placebo SD	Perm. p	DID (pp)	Placebo SD	Perm. p
AR[0]	0.857***	0.304	0.002	1.852***	0.759	0.010
CAR[-1,+1]	0.606*	0.363	0.093	1.446*	0.771	0.059
CAR[-2,+2]	0.919**	0.373	0.011	1.912**	0.842	0.022

*Notes: Placebo test based on 10,000 random permutations of term labels. The DID is computed as mean |CAR|_Term2 − mean |CAR|_Term1 for SPY only, and as the cross-market difference-in-differences for the US vs CN panel. "Placebo SD" is the standard deviation of the permuted DID distribution. "Perm. p" is the two-sided permutation p-value. The actual DID falls in the extreme tail of the placebo distribution at the event-day window, indicating that the observed cross-term amplification is unlikely to arise by chance. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*